

Jumarald Pharmacy & Wellness

Full AI Architecture & Security CIA Audit Documentation — v3.0.0

1. Executive Summary

Jumarald Pharmacy & Wellness is a digital healthcare platform operating in Ghana. The system unifies e-commerce pharmaceutical sales, certified prescription validation, telehealth physician appointments, real-time multi-branch inventory tracking, and an intelligent **Superintendent Clinical AI Assistant ('Dr. Jumarald AI')**.

Superintendent Pharmacist: Pharm. Philip Bruce-Tagoe (GPHC Reg. No. 2050984).

2. Superintendent AI Assistant Architecture ('Dr. Jumarald AI')

Dr. Jumarald AI features a Dual-Provider Factory (OpenAI GPT-4o-mini / Gemini 1.5 Flash), an autonomous Intent Classifier, Tool Registry, and Rule-Based Fallback Engine:

- **Intent Classifier Pipeline:** PRODUCT_SEARCH, DRUG_INTERACTION, PRESCRIPTION_ANALYSIS, BRANCH_INQUIRY, SYMPTOM_Triage, GENERAL_CONSULTATION.
- **Registered AI Tools:** searchProducts, getProductDetails, checkDrugInteraction, findNearbyBranches, getUserOrders, createPharmacistConsultation.
- **Clinical Fallback Engine:** Zero-downtime safety net triggering emergency warnings (Ghana 112 / 193).

3. Comprehensive Security & CIA Triad Audit

3.1 Confidentiality Audit: Stateless JWT sessions, bcrypt password hashing, active user DB checks in authenticateToken middleware, encrypted Cloudinary streams.

3.2 Integrity Audit: Prisma ACID transactions for inventory updates, Zod input validation, Paystack HMAC SHA-512 webhook verification, AuditLog tracking.

3.3 Availability Audit: Dedicated rate limiters (authLimiter: 20/15m, aiLimiter: 20/m, prescriptionLimiter: 30/15m), domain-restricted CORS, Socket.io auto-invalidation.

4. Integrated External APIs & Services

- **Paystack Payment API:** Ghana Mobile Money (MoMo) & Card checkout in GH₵ with HMAC SHA-512 webhooks.
- **OpenAI GPT-4o-mini & Gemini 1.5 Flash APIs:** Core LLM engines powering Dr. Jumarald AI.
- **Cloudinary Cloud CDN API:** Encrypted prescription uploads & optimized product images.
- **Nodemailer / SMTP API:** Transactional emails for receipts, prescription verification, & password resets.
- **WebPush / VAPID API:** Real-time browser notifications for rider tracking.
- **Sentry Telemetry API:** Application performance monitoring and backend error tracking.

5. Strategic Future Software Procurement Roadmap

- **Twilio / Hubtel SMS Gateway API (Paid Service):** Direct SMS notifications for Ghanaian mobile numbers (OTP logins, instant prescription refill alerts, rider ETA).

- **Upstash Redis Cluster (Paid Subscription):** High-speed distributed in-memory session caching & Socket.io pub/sub synchronization.
- **AWS S3 / Cloudflare R2 Vault (Paid Cloud):** HIPAA & Ghana DPC compliant long-term encrypted prescription archiving.
- **RxNorm / NLM Drug Interaction API (License):** Global pharmaceutical API integration for automated contraindication scanning.
- **IdentityPass Ghana Card Verification API (Paid Service):** Automated identity validation for patients and doctors.
- **Google Maps Platform API (Paid License):** Precise distance calculation, rider geolocation tracking, & dynamic delivery fees.